



Facial Attribute Detection

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Group 27

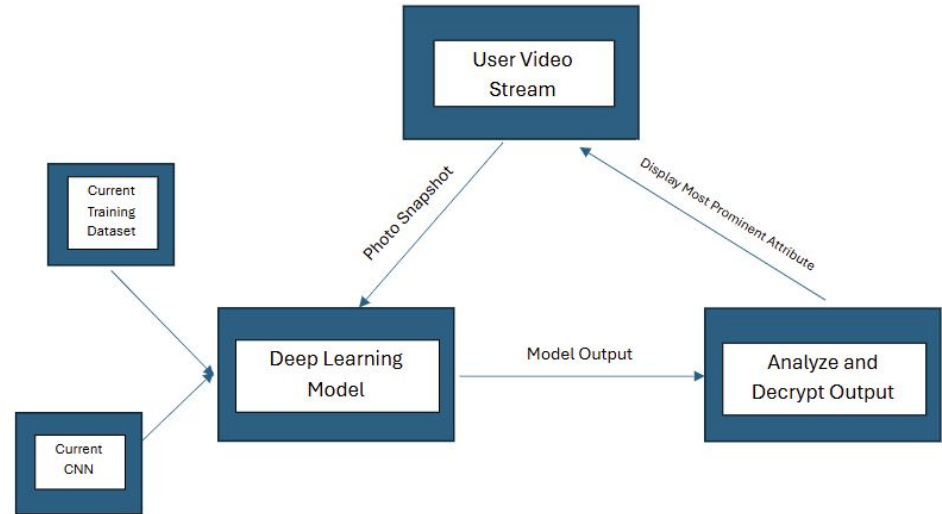
Introduction

- Facial attribute detection is a powerful tool powered by CNN's that identifies key features on a human face, unlocking a wide range of applications
- Objective of detecting faces as well as identifying 20 facial features in a given dataset
- Integrated with real-time detection



Approach

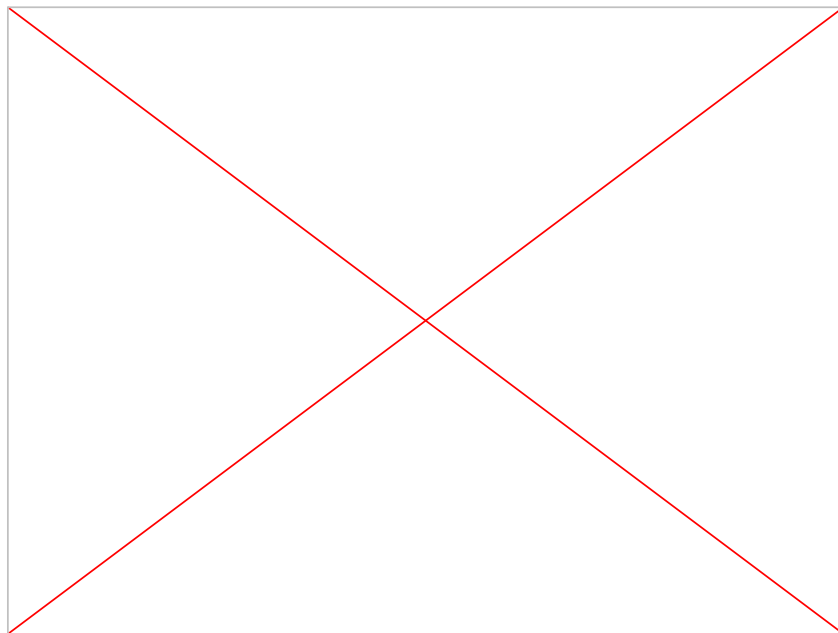
- **Face Detection & Feature Extraction:** The project utilizes MTCNN and Multi-CNN models to detect faces and identify facial features in live video streams and datasets, inspired by a landmark study on MTCNN's superior performance in face detection.
- **MTCNN Process:** It operates through three stages — Proposal Network (P-Net) for rough face regions, Refinement Network (R-Net) for better precision, and Output Network (O-Net) for final bounding boxes and facial landmarks.
- **Attribute Recognition:** Uses CNNs to identify global and specific facial attributes, employing convolutional layers, pooling, and classification networks for detailed feature analysis.





Real-World Applications

- **Cosmetology and Personalization:** Tailoring beauty and skincare recommendations based on individual facial attributes
- **Security and Surveillance:** Enhancing identity verification, emotion recognition, and monitoring systems through facial analysis
- **Healthcare and Diagnostics:** Assisting in diagnosing genetic conditions or analyzing facial features for early detection of diseases





Challenges and Ethical Considerations

Challenges

- Optimization of MTCNN layers by uning hyperparameters and balancing detection accuracy with computational efficiency
- Training on datasets that lack diversity can lead to biases in detection accuracy across different demographics

Ethical Considerations:

- Handling and storing facial data must comply with regulations to protect user privacy.



Conclusion & Call to Action

- Successfully implemented an MTCNN-based model for real-time face detection and identification of distinct facial features, showcasing potential applications in fields like cosmetology and personalized services.
- Aim to enhance accuracy and robustness through larger datasets, optimized CNN layers, and alternative approaches like MMCNN for better performance.
- Stay curious, stay informed, and explore how you can be part of this exciting journey!